



AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE

Charlie

FINAL SCORE

21 / 70

weighted points

CORRECT ANSWERS

11 / 35

questions

NEEDS IMPROVEMENT

Assessment Date: July 24, 2026

Report Generated: July 24, 2026 at 21:44 UTC

Role: Data Science Expert

OVERALL RATING

★ ★ ★ ★ ★

SUCCESS RATE

30.0%

WEIGHTED SCORE

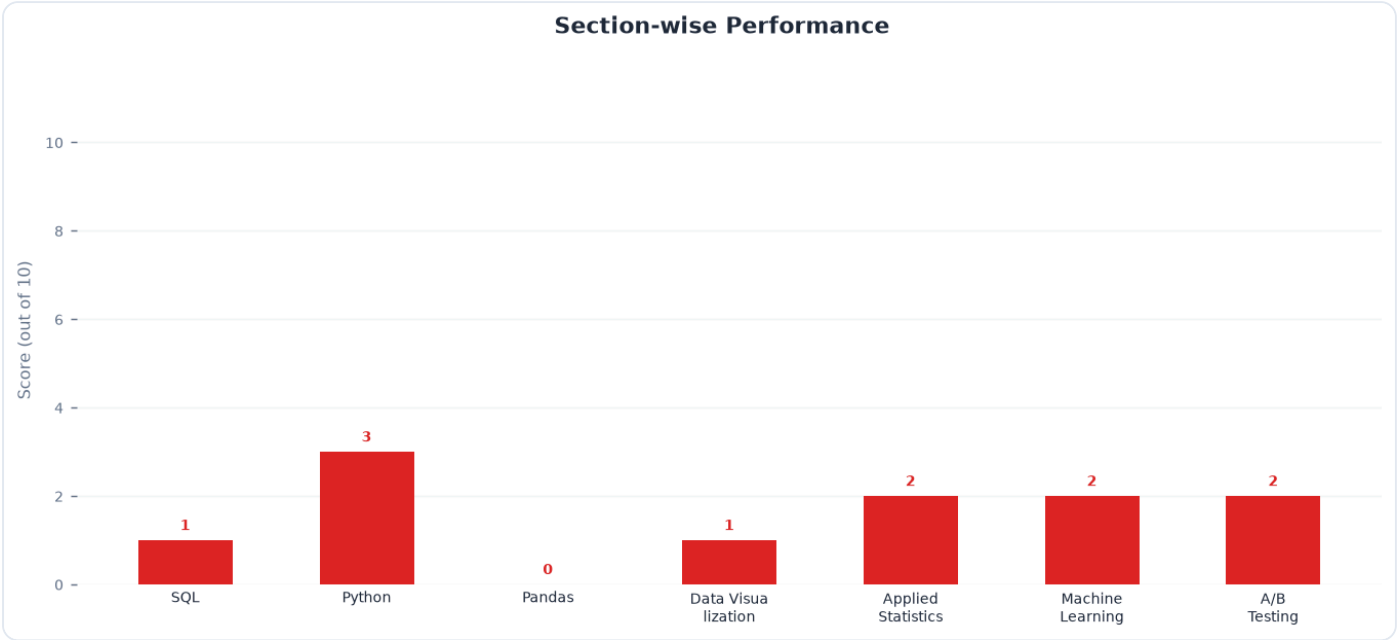
21 / 70

TOP SKILL

Python

NEEDS IMMEDIATE ATTENTION

Pandas



Skill Rating & Benchmark

Section	Your Score	Reference Benchmark (Demo)	Delta	Status
SQL	1 / 10	6.8	-5.8	Critical Gap
Python	3 / 10	6.2	-3.2	Needs Work
Pandas	0 / 10	5.8	-5.8	Critical Gap
Data Visualization	1 / 10	6.0	-5.0	Critical Gap
Applied Statistics	2 / 10	5.5	-3.5	Needs Work
Machine Learning	2 / 10	5.2	-3.2	Needs Work
A/B Testing	2 / 10	6.4	-4.4	Critical Gap

Attempt Statistics Breakdown

Section	Total Q	Attempted	Correct	Incorrect	Skipped	Accuracy
SQL	5	5	1	4	0	20%
Python	5	5	3	2	0	60%
Pandas	5	5	0	5	0	0%
Data Visualization	5	5	1	4	0	20%
Applied Statistics	5	5	2	3	0	40%
Machine Learning	5	5	2	3	0	40%
A/B Testing	5	5	2	3	0	40%

Executive Summary

The candidate achieved a correct-answer count of 11 out of 35 (31.4% accuracy) with a weighted score of 21 out of 70, indicating critical gaps across core data science competencies. While demonstrating basic programming familiarity, the candidate struggled with intermediate-to-advanced SQL, pandas operations, statistical diagnostics, and rigorous A/B testing methodologies. Targeted technical upskilling is highly recommended to meet industry-standard expectations for senior data science roles.

Key Strengths

- Foundational Python Syntax:** Demonstrates a baseline understanding of core Python structures and basic programming logic.
- Core Machine Learning Concepts:** Shows familiarity with standard supervised learning workflows and model training

- **Basic Data Visualization:** Understands the primary purposes of standard plotting libraries and basic exploratory data analysis.
- **General Statistical Awareness:** Recognizes fundamental statistical terms and elementary probability concepts.

Top Blind Spots

- **Advanced SQL & Pandas Operations:** Gaps exist in understanding pandas-specific index behaviors (such as `.loc` inclusivity and monotonic constraints in rolling windows) and advanced SQL constructs like recursive CTEs and date truncation.
- **Statistical Diagnostics & Experimental Design:** The candidate struggled to diagnose regression violations like heteroscedasticity, calculate standard error, and apply advanced A/B testing techniques like the Delta Method for ratio metrics.
- **ML Evaluation & Feature Engineering:** Gaps were identified in understanding how feature scaling impacts distance-based algorithms (like k-NN) and selecting appropriate metrics (PR-AUC vs. ROC-AUC) for highly imbalanced datasets.

Actionable Learning Resources

SQL & Pandas: Practice advanced window functions and complex joins on StrataScratch, and review the pandas documentation on time-series offsets and monotonic indexes.

✓ Action Item

Statistical Modeling: Read Chapter 3 of *An Introduction to Statistical Learning (ISLR)*, focusing on OLS assumptions, residual analysis, and variance-stabilizing transformations.

✓ Action Item

A/B Testing: Study *Trustworthy Online Controlled Experiments* by Kohavi et al., specifically focusing on statistical power, one-sided vs. two-sided tests, and the Delta Method for ratio metrics.

✓ Action Item

Data Visualization: Review Matplotlib's official performance optimization guides, focusing on handling large datasets (10^6 + points) using `rasterized=True`.

✓ Action Item